Name: _____

TOPIC TEST 2

Chapter 4: Recurrence relations

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 6 short-answer questions (15 marks)
- Part C: 1 analysis question (15 marks)

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 The first four terms of a sequence are given below.

4, 6, 12, 30

A recurrence relation that can be used to generate this sequence is:

A $t_{n+1} = 2t_n - 2$ $t_1 = 4$

B $t_{n+1} = 3t_n - 6$ $t_1 = 4$

C $t_{n+1} = 1.5t_n$ $t_1 = 4$

D $t_{n+1} = t_n + 2$ $t_1 = 4$

E $t_n = t_{n-1} + t_{n-2} + 2$ $t_1 = 4, t_2 = 6$

[1 mark]

2 A recurrence relation for a sequence is $w_n = 3w_{n-1} + 12$. If $w_5 = 966$, then w_3 is equal to:

A 318

B 45.375

C 30

D 8742

E 102

[1 mark]

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3 A sequence is described by the recurrence relation $t_{n+1} = t_n + 7$ where $t_1 = 13$. The nth term of this sequence is given by:

A $t_n = 7n + 6$

B $t_n = 13 + 7n$

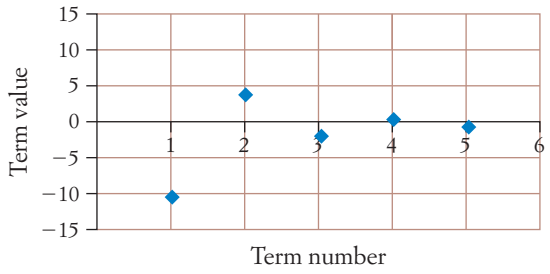
C $t_n = 7 + n$

D $t_n = 13 + 3n$

E $t_n = 14 + 7n$

[1 mark]

4



The values of the first five terms of a sequence are plotted on the graph above. A recurrence relation that could apply to this sequence is:

A $t_{n+1} = 0.4t_n$ $t_1 = -10$

B $t_{n+1} = -2.5t_n$ $t_1 = 10$

C $t_{n+1} = -2.5t_n$ $t_1 = -10$

D $t_{n+1} = -0.4t_n$ $t_1 = -10$

E $t_{n+1} = 20 - 2.5t_n$ $t_1 = 10$

[1 mark]

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5 A new plant, standing 25 cm tall, is added to a garden. During its first month, its height increased by 12.5 centimetres.

During its nth month since being planted, the height of the plant increased by H_n centimetres where $H_{n+1} = 0.9H_n$.

The maximum height to which the plant can grow, in centimetres, is closest to:

A 150

B 39

C 125

D 250

E 25

[1 mark]

Part B

Short-answer questions

6 short-answer questions
15 marks
Show your working where appropriate.

1 Write down the first five terms of the sequence defined by the recurrence relation:

$$t_{n+1} = 3t_n + 6 \qquad t_1 = -4$$

[2 marks]

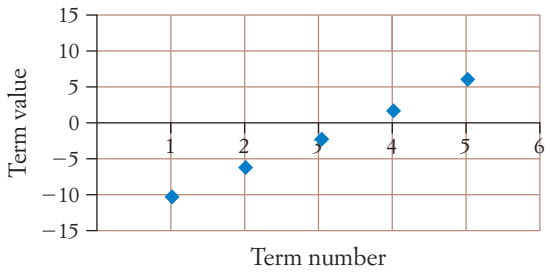
2 A sequence is defined by the rule $t_n = 100 \times 0.8^{n-1}$, $n = 1, 2, 3, \ldots$

Write a recurrence relation that describes the same sequence.

[2 marks]

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3 The graph below shows the first five terms of a recurrence relation.



a Write the recurrence relation.

b What is the gradient of the line that passes through these five points?

[3 marks]

4 The nth term of an arithmetic sequence is given by the rule:

$$t_n = 10 - 4n \qquad n = 1, 2, 3, \ldots$$

Find the recurrence relation.

[2 marks]

5 A population of bandicoots decreases in size by 15% each year. There are initially 500 bandicoots in the population.

a Find a recurrence relation that represents the size of the bandicoot population each year.

b Once the bandicoot population reaches 50 individuals, the population will no longer be able to sustain itself and they will be taken into a program for protection. After how many years will this happen?

[3 marks]

6 Andrew invests $6000 in a savings account that earns 7.5% per annum simple interest.

a How much interest is earned each year?

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b Find a recurrence relation that gives the amount in Andrew's savings account after n years.

[3 marks]

Part C

Analysis question

1 analysis question
15 marks
Show your working where appropriate.

1 Zoe opens an online gift shop. In the first month, she sells 18 pairs of handmade earrings from her stock. In the second month, 25 pairs were sold and in the third month, 32 pairs of earrings were sold.

The number of earring pairs sold each month form an arithmetic sequence:

18, 25, 32, …

a A rule describing this sequence is $E_{n+1} = E_n + x$ where $E_1 = 18$. State the value of x.

b How many pairs of earrings were sold in the fifth month after opening?

c How many more pairs of earrings were sold in the sixth month than in the fourth month?

d i How many pairs of earrings were sold in the first 6 months after opening?

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ii If Zoe can make a maximum of 100 earrings per month, how many more months until she reaches this limit?

In the first month after Zoe's gift shop opens, she receives 24 orders for custom necklaces. After the first month, the number of custom necklace orders each month forms a geometric sequence with a common ratio of 1.5.

The sequence that describes the number of custom necklace orders in the nth month after the shop opens can be written as the recurrence relation:

$$C_{n+1} = aC_n + b \qquad C_1 = k$$

e Write the values of a, b and k.

f How many orders for the necklaces did the store receive in the third month after opening?

g How many custom necklaces were ordered in total in the first 4 months after opening?

Zoe uses the following recurrence relation to model the profit, in dollars, P_n, made by her online gift store at the end of the nth year after opening.

$$P_{n+1} = 0.9P_n + 3000 \qquad P_1 = 8500$$

h What is the profit made at the end of the third year after the shop opens?

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i At the end of which year after the shop opens does the profit first exceed $20 000?

j According to this model, what is the largest possible profit that can be made in a year?

[15 marks]

[Total marks: 35]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 The first term is 4, so $t_1 = 4$.

Each term in the sequence is 6 less than three times the previous term.

$3 \times 4 - 6 = 6$

$3 \times 6 - 6 = 12$

$3 \times 12 - 6 = 30$

$\therefore t_{n+1} = 3t_n - 6 \qquad t_1 = 4$

$\therefore$ **B**

2 $w_5 = 3w_4 + 12$

$966 = 3w_4 + 12$

$954 = 3w_4$

$W_4 = 318$

$W_4 = 3w_3 + 12$

$318 = 3w_3 + 12$

$306 = 3w_3$

$W_3 = 102$

$\therefore$ **E**

3 The recurrence relation represents an arithmetic sequence with $a = 13$ and $d = 7$.

$t_n = a + (n - 1)d$

$= 13 + 7(n - 1)$

$= 13 + 7n - 7$

$= 6 + 7n$

$\therefore$ **A**

4 The sequence is geometric with $t_1 = -100$ and $r = -0.4$ which is most easily found by $\frac{t_2}{t_1} = \frac{4}{-10}$.

The recurrence relation is therefore $t_{n+1} = -0.4t_n \qquad t_1 = -100$

$\therefore$ **D**

5 The growth sequence is geometric with $r = 0.9$ and $a = 12.5$.

$S_\infty = \frac{a}{1 - r}$

$= \frac{12.5}{1 - 0.9}$

$= 125$

The maximum possible height of the plant is $25 + 125 = 150$ cm.

$\therefore$ **A**

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Part B – Short-answer questions

1 $-4, -6, -12, -30, -84$ **[1 mark for two correct, 2 marks for five correct]**

2 $t_{n+1} = 0.8t_n$ **[1 mark]** $t_1 = 100 \times 0.8^0 = 100$ **[1 mark]**

3 a $t_{n+1} = t_n + 4$ **[1 mark]** $t_1 = -10$ **[1 mark]**

b Gradient $= 4$ **[1 mark]**

4 $t_{n+1} = t_n - 4$ **[1 mark]** $t_1 = 6$ **[1 mark]**

5 a $t_n = 0.85t_{n-1}$ **[1 mark]** $t_0 = 500$ **[1 mark]**

b $t_{15} = 51.3848$

$t_{16} = 43.6771$

In the 16th year, the bandicoot population will be taken into a program for protection. **[1 mark]**

6 a $\frac{7.5}{100} \times 6000 = 450$

Each year, \$450 is earned in interest. **[1 mark]**

b $t_n = t_{n-1} + 450$ **[1 mark]** $t_0 = 6000$ **[1 mark]**

Part C – Analysis question

1 a A rule describing this sequence is $E_{n+1} = E_n + x$ where $E_1 = 18$. State the value of x.

$x = 7$ **[1 mark]**

b $E_4 = 32 + 7$

$= 39$

$E_5 = 39 + 7$

$= 46$ **[1 mark]**

c Seven more pairs of earrings are sold in the fifth month than in the fourth month, and seven more pairs are sold in the sixth month than in the fifth month .

Therefore, 14 more pairs of earrings are sold in the sixth month than in the fourth month. **[1 mark]**

d i $a = 18, d = 7$

$S_n = \frac{n}{2}[2a + (n-1)d]$

$S_6 = \frac{6}{2}[2 \times 18 + 5 \times 7]$

$= 213$ **[1 mark]**

ii $t_{12} = 95$

$t_{13} = 102$

Zoe will reach her limit of making 100 earrings per month in the 13th month. **[1 mark]**

This is 7 months later. **[1 mark]**

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e $a = 1.5$ **[1 mark]**

$b = 0$ **[1 mark]**

$k = 24$ **[1 mark]**

f $C_2 = 1.5 \times 24$

$= 36$

$C_3 = 1.5 \times 36$

$= 54$

$\therefore$ The store received 54 orders for custom necklaces in the third month after opening. **[1 mark]**

g $C_4 = 1.5 \times 54$

$= 81$

Total necklaces ordered in the first 4 months after opening

$= 26 + 36 + 54 + 81$

$= 195$ **[1 mark]**

h $P_2 = 0.9 \times 8500 + 3000$

$= 10\,650$

$P_3 = 0.9 \times 10\,650 + 3000$

$= 12\,585$

The profit made at the end of the third year is \$12 585. **[1 mark]**

i At the end of the 9th year, the profit is \$20 744.95. So, at the end of the 9th year, the profit first exceeds \$20 000. **[1 mark]**

j Let x be the maximum possible profit that can be made in a year.

$P_{n+1} = 0.9P_n + 3000$

$x = 0.9x + 3000$ **[1 mark]**

$x = 30\,000$

The maximum possible profit that can be made in a year is \$30 000. **[1 mark]**

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